Hard & Soft palate

Palate

- **Forms the arched** roof of the mouth and floor of the nasal cavities.
- separates the oral cavity from the nasal cavities and the nasopharynx.

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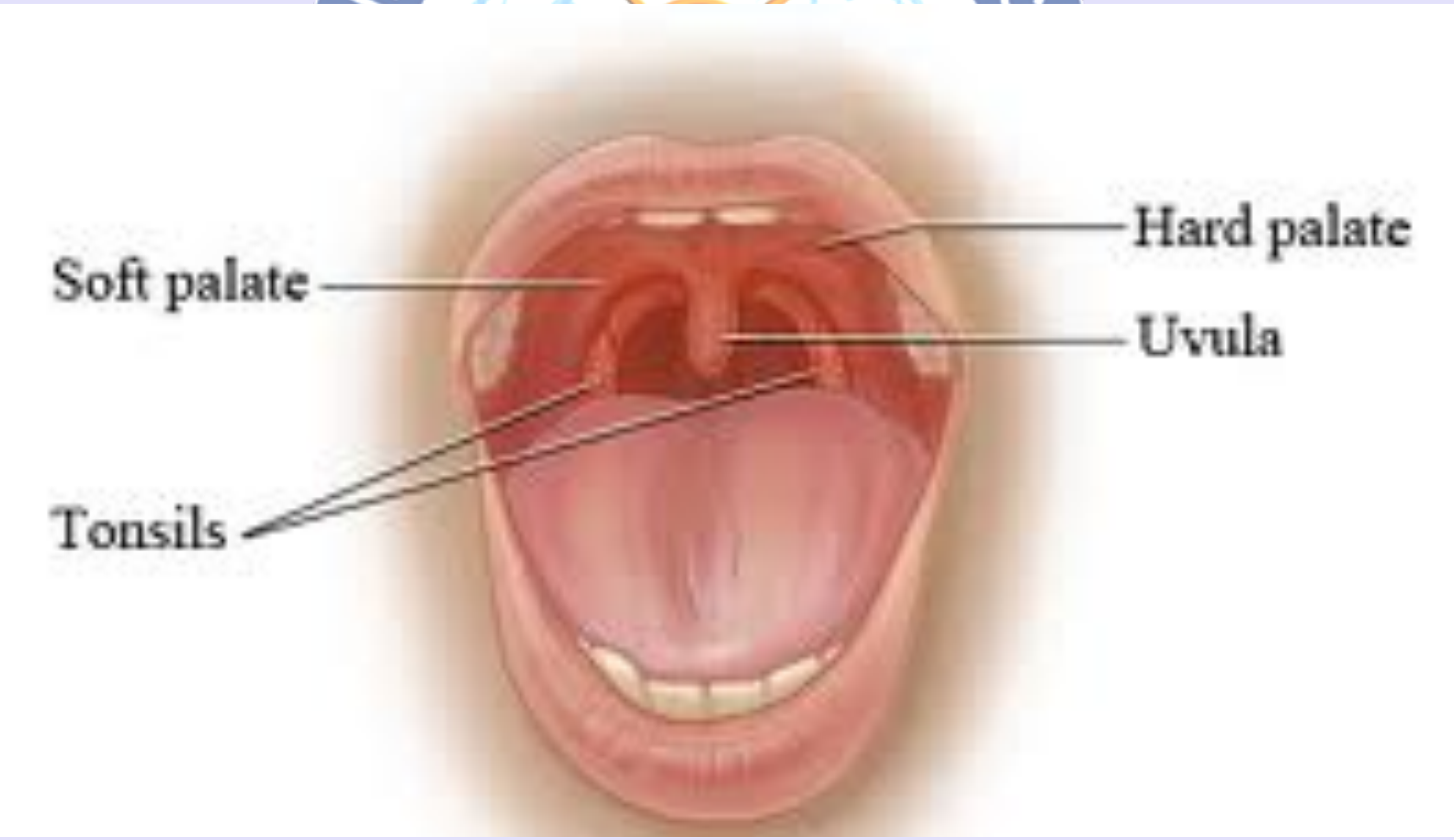
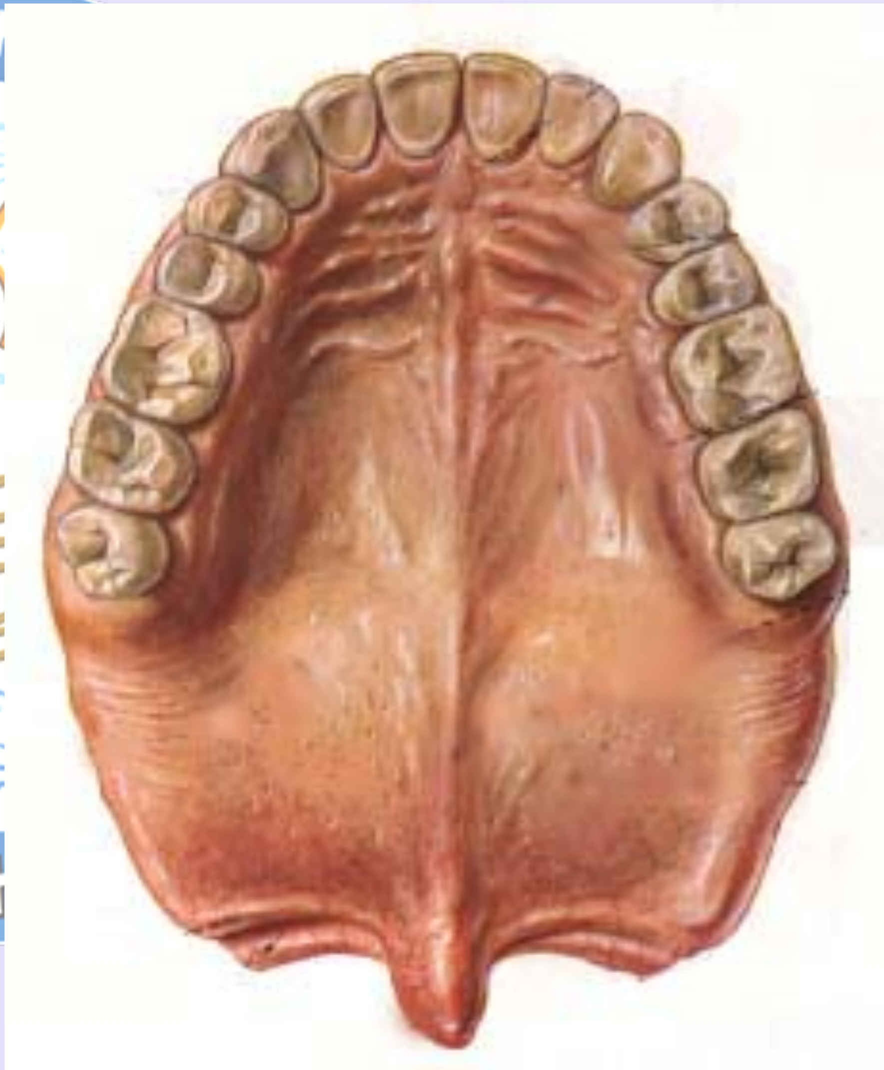
Palate

- Palate is divided into two parts:
  - Anterior bony **hard palate**
  - Posterior fleshy **soft palate** or **velum**



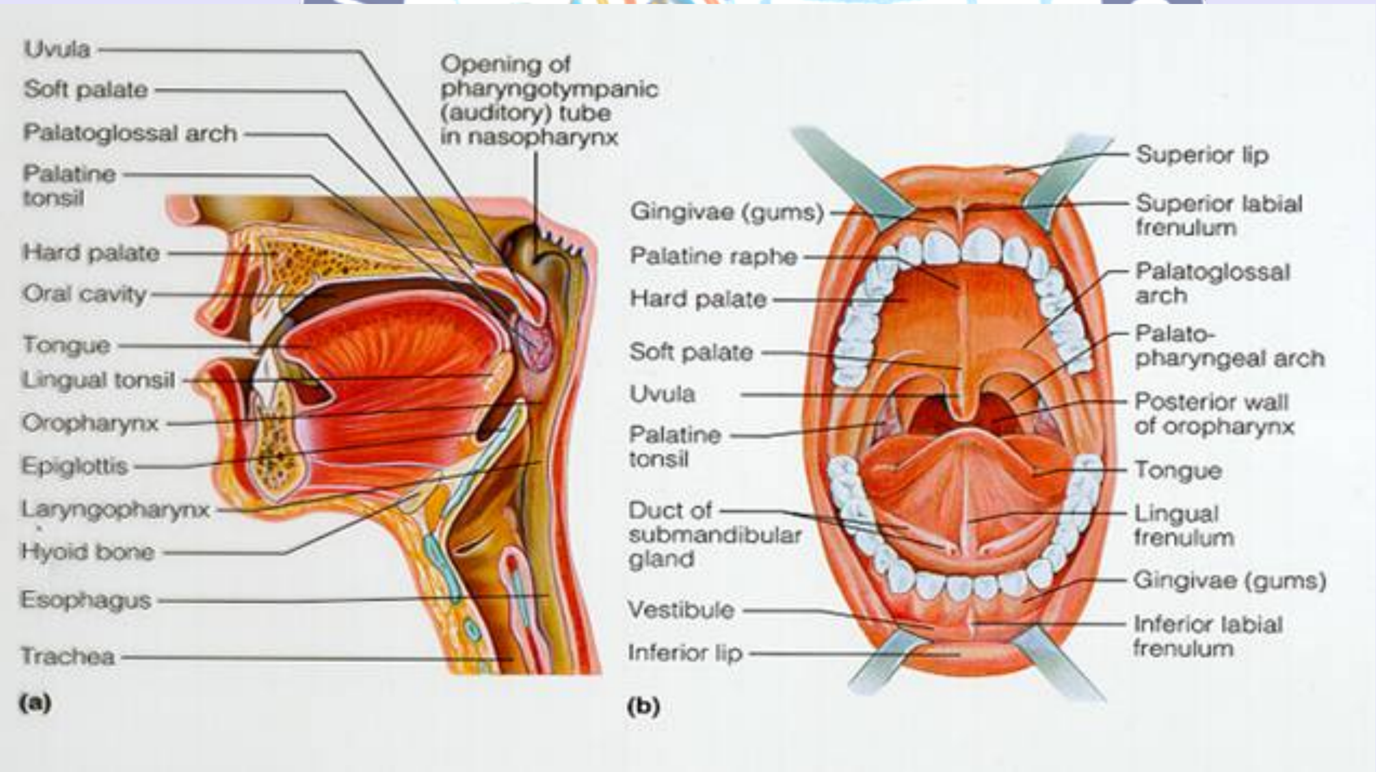
Hard palate

- **Vaulted (concave)** , space mostly filled by tongue at rest.
- Anterior two thirds of the palate has a bony skeleton formed by the palatine process of the maxilla and horizontal plate of palatine bone.



Hard palate

- forms a partition between nasal passages and mouth. This partition is continued deeper into the mouth by a fleshy extension called the soft palate.

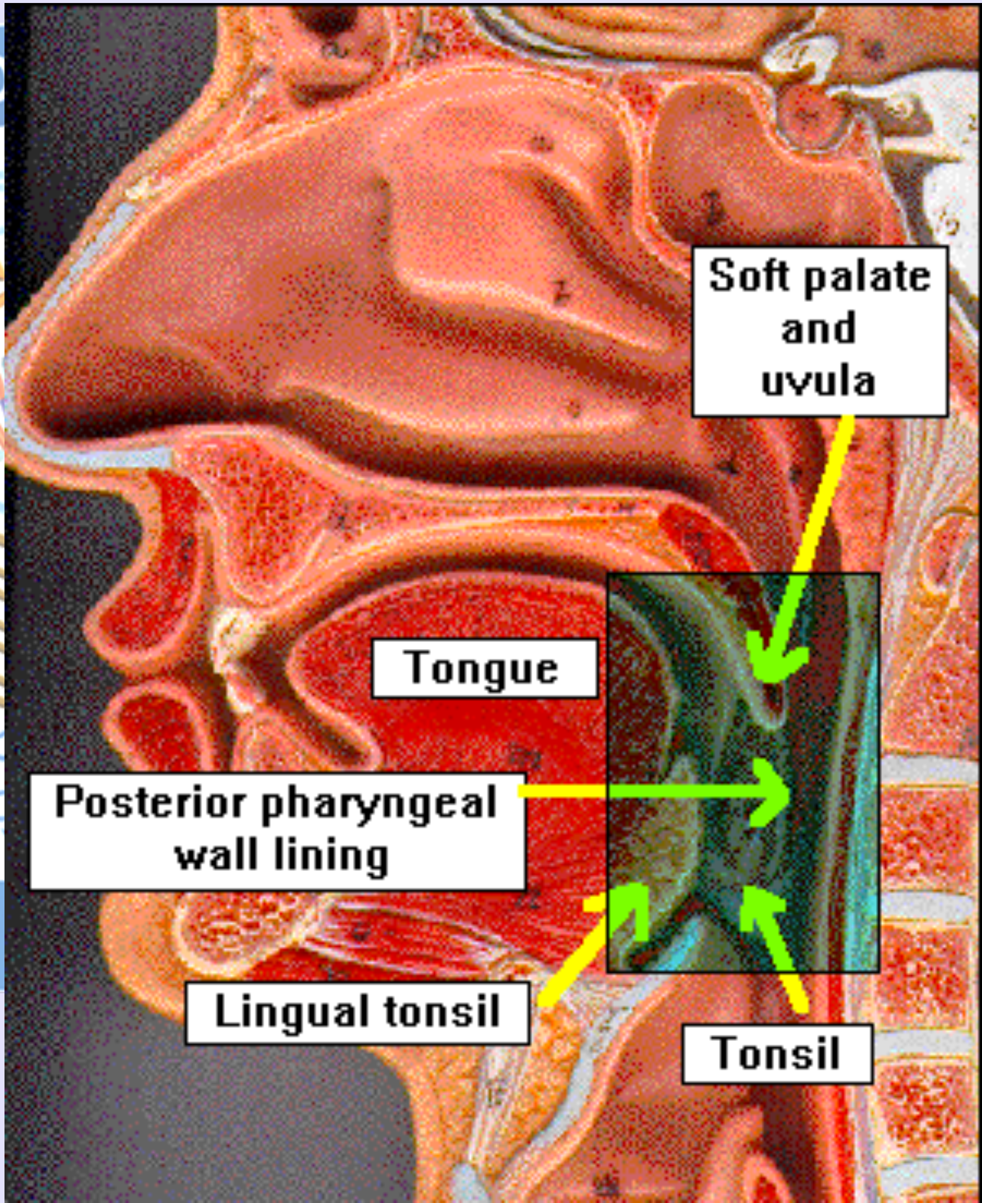


Function Of the hard palate

- Interaction between the tongue and the hard palate is essential in the formation of certain speech sounds, notably t, d, j

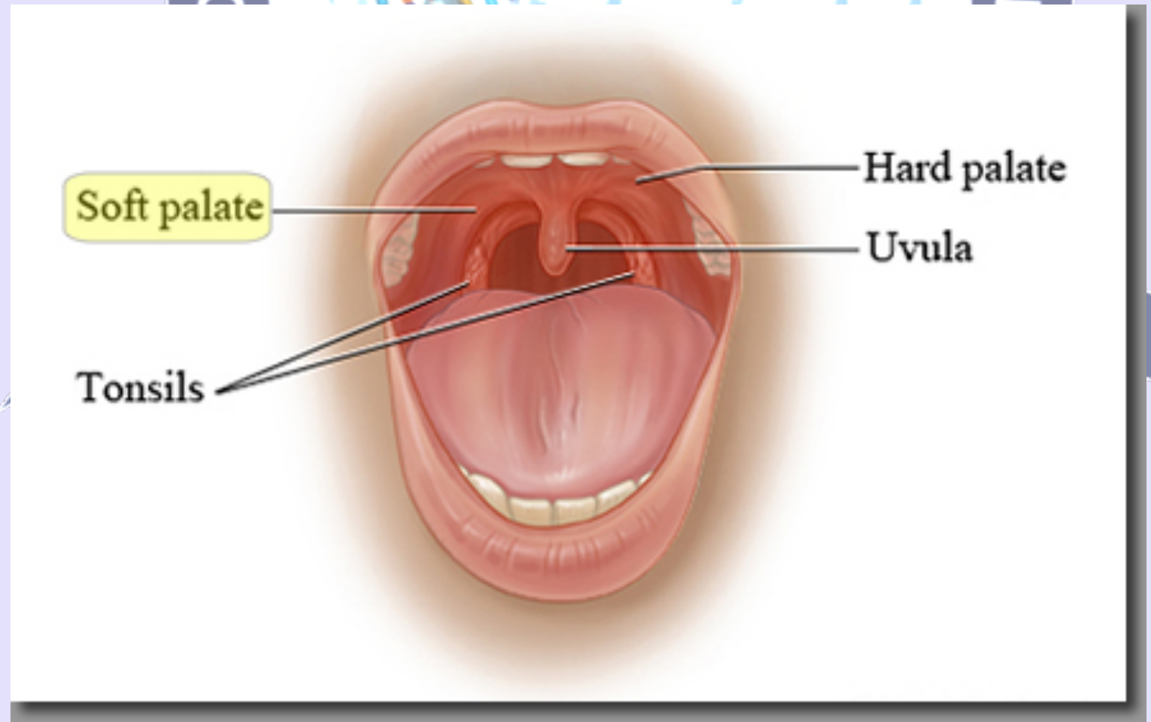
Soft palate

- Also known as **velum** or **muscular palate**
- Soft tissue constituting back of the roof of the mouth.
- Distinguished from the hard palate at front of mouth in that it does not contain bone.



Functions of soft palate

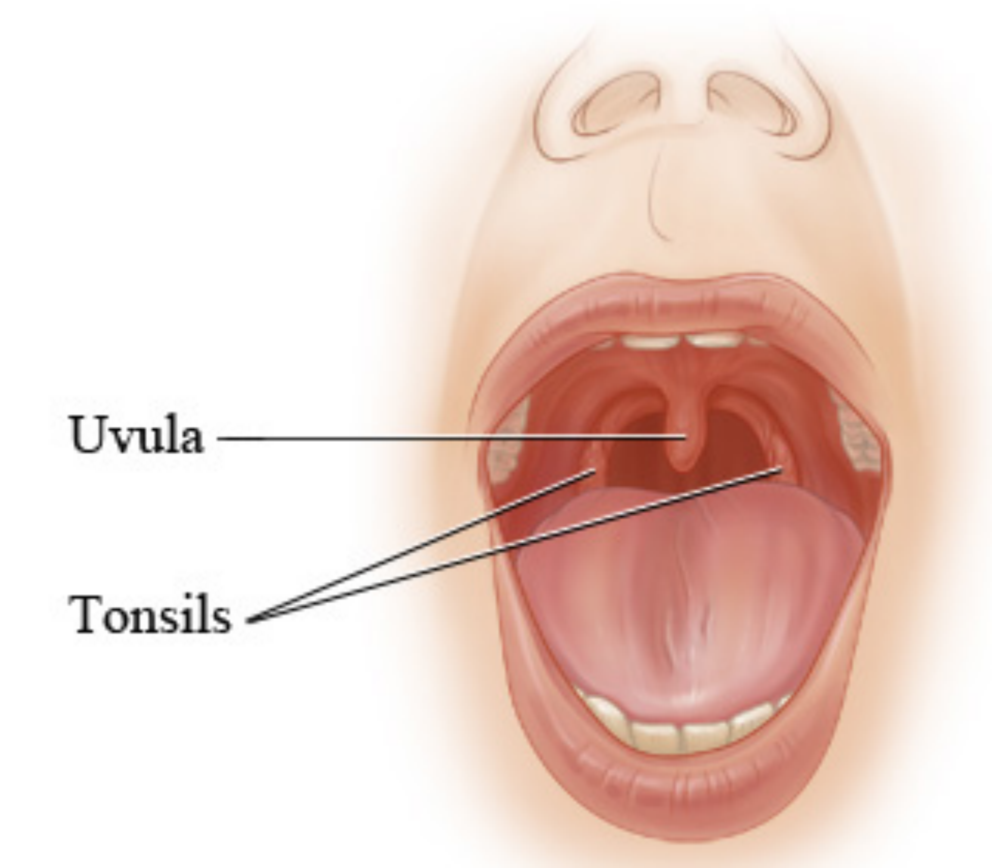
- Soft palate is **movable**.
- Consisting of muscle fibers sheathed in mucous membrane.
- Closes off the nasal passages during the act of swallowing,& for closing off the airway.



Functions of soft palate

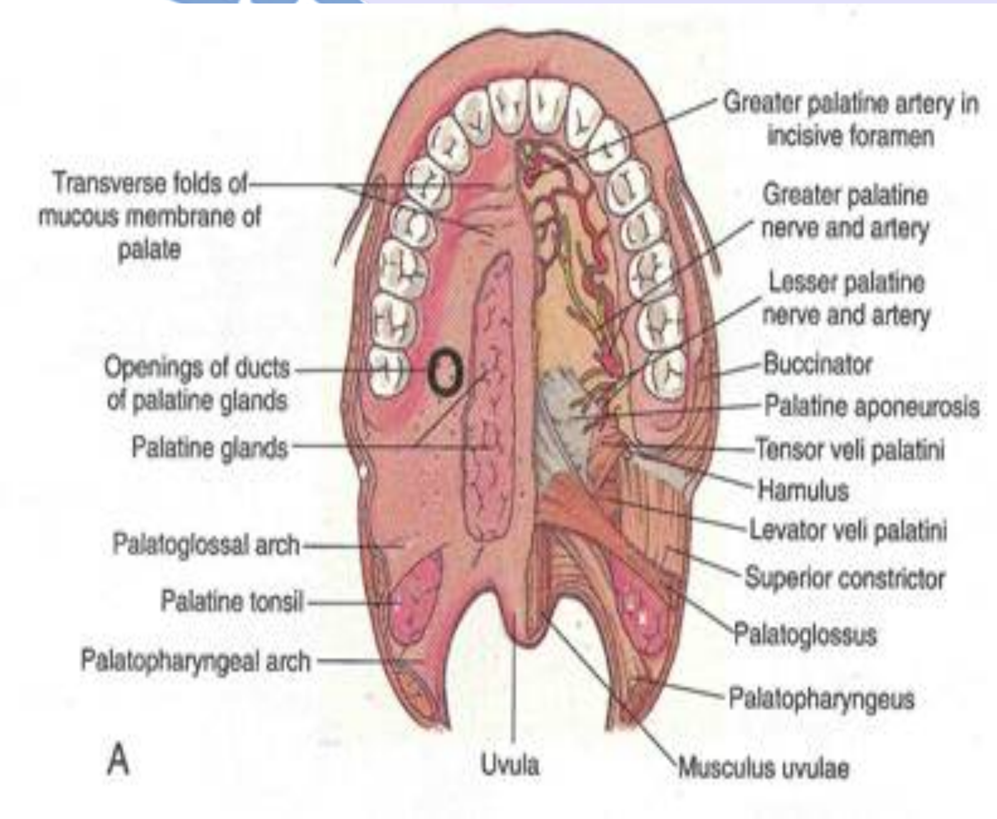
- Protects the nasal passage by diverting a portion of the excreted substance to the mouth during sneezing.

- **Uvula** hangs from the end of soft palate.



Vasculature Of Palate

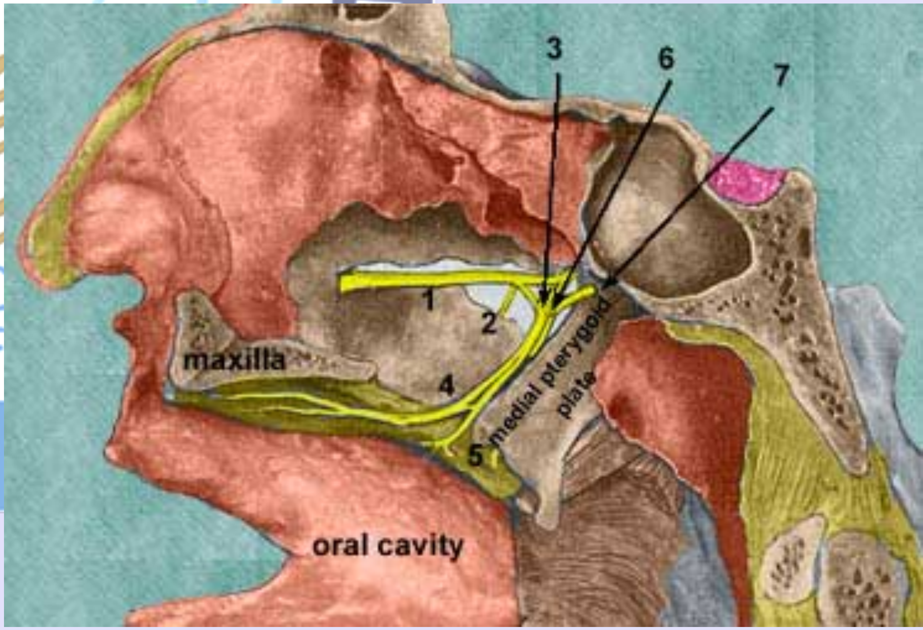
- Palate has a **rich** blood supply.
- Chiefly from **the greater palatine artery** on each side, a branch of the descending palatine artery..
- Veins of the Palate are tributaries of the **pterygoid venous plexus**.



Innervation of the palate

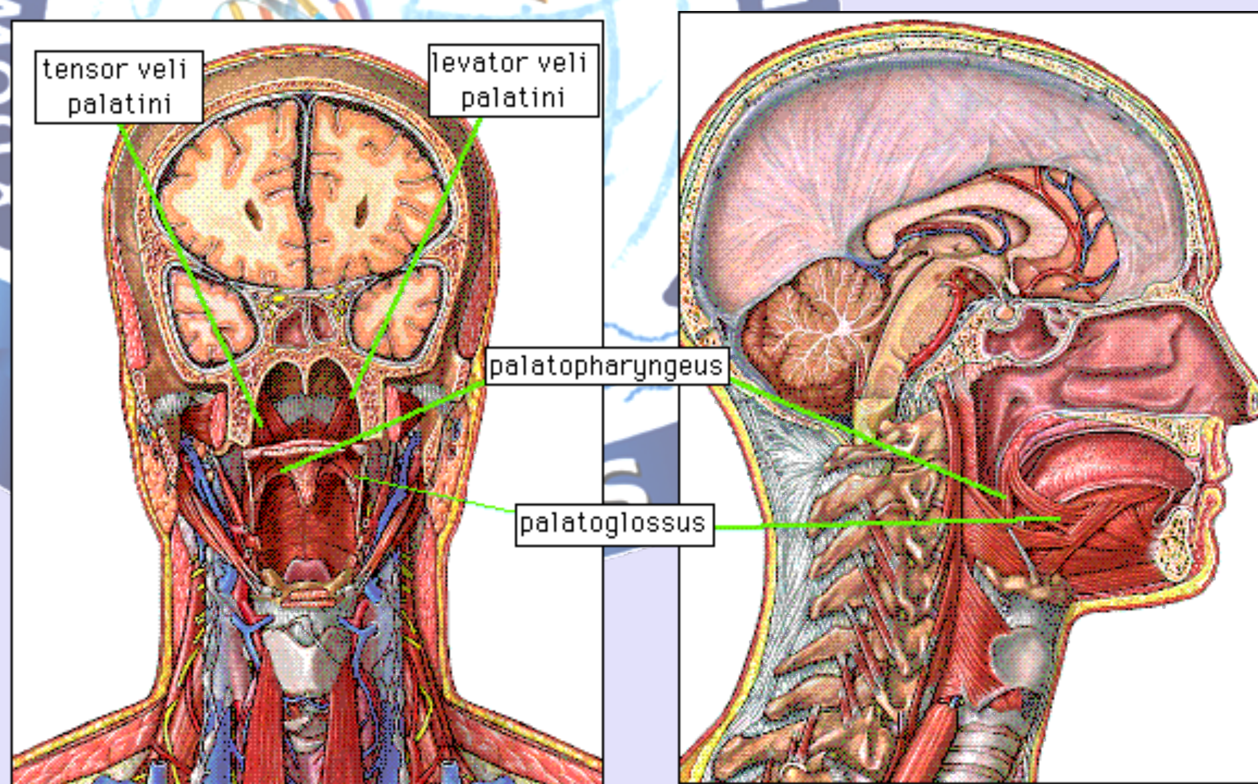
- **Sensory nerves of the palate are branches of the maxillary nerve, which branch from the pterygopalatine ganglion.**
- **Greater palatine nerve supplies gingiva, mucous membrane and glands of most of hard palate.**
- **Nasopalatine nerve supplies mucous membrane of ant part of hard palate.**
- **Lesser palatine supply soft palate.**

3.pterygopalatine ganglion (parasympathetic)
4.greater palatine nerve
5.lesser palatine nerve



Muscles of soft palate

- Muscle of the soft palate play important roles in **swallowing** and **breathing.**
- **Five muscles** of the soft palate arise from the base of the cranium and descend to the palate.



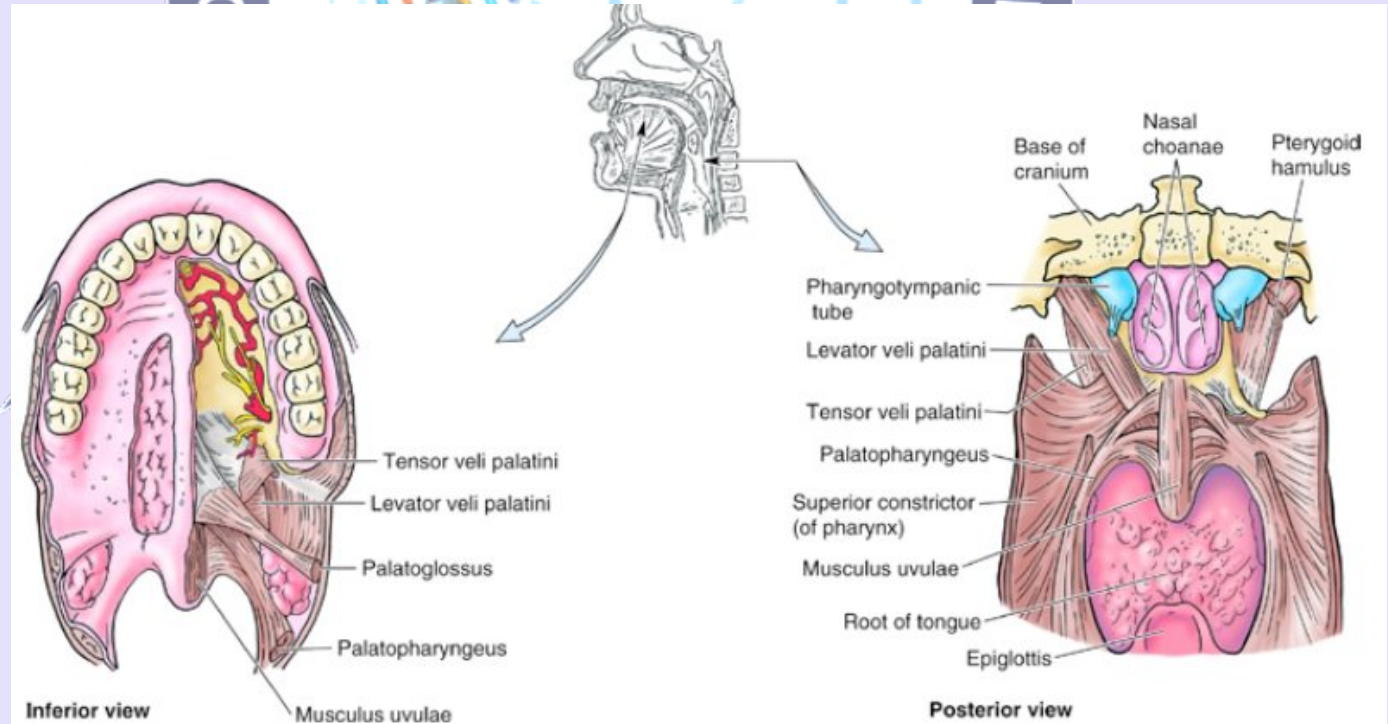
1.Levator veli palatini Attachment

Superior Attachment:

- Cartilage of pharyngotympanic tube.
- Petrous part of temporal bone.

Inferior Attachment:

- Palatine Aponeurosis.



Levator Veli Palatini Innervation

- Innervated via the pharyngeal plexus, primarily by the pharyngeal branch of the **vagus nerve (CN X).**

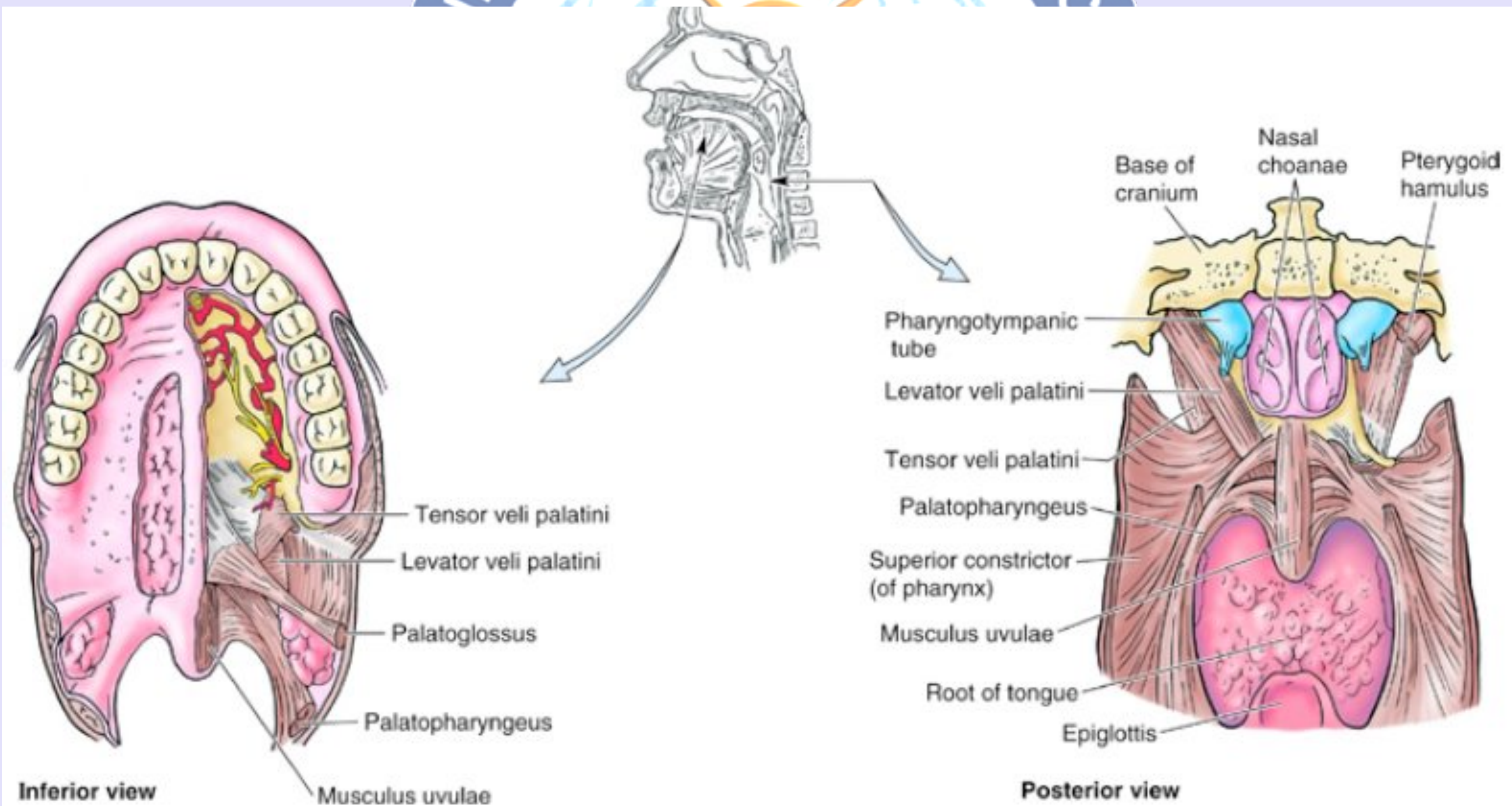
Levator Veli Palatini

Action:

- **Elevator muscle** of the **soft palate during swallowing and yawning**..
- During swallowing, it contracts, elevating the soft palate to help prevent food from entering the nasopharynx.

2. Tensor Veli Palatini

A broad, thin, ribbon-like muscle in the head that tenses the soft palate.



Attachment

Superior Attachment:

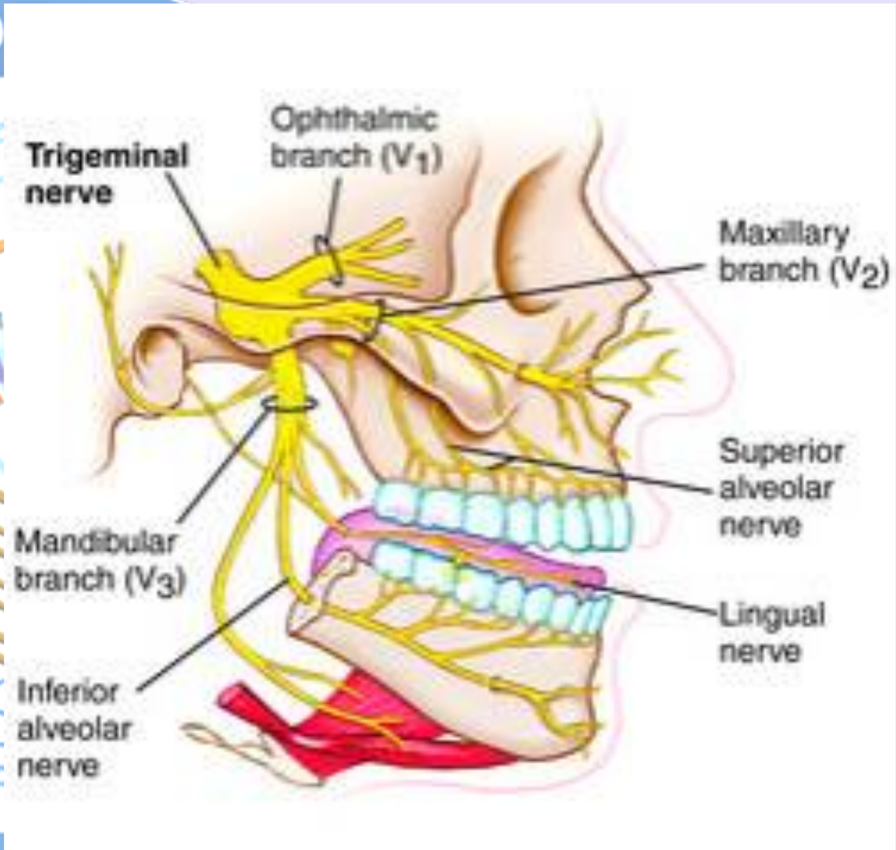
- Scaphoid fossa of medial pterigoid plate,
- Spine of sphenoid bone,
- Cartilage of pharyngotympanic tube.

Inferior Attachment:

Palatine Aponeurosis.

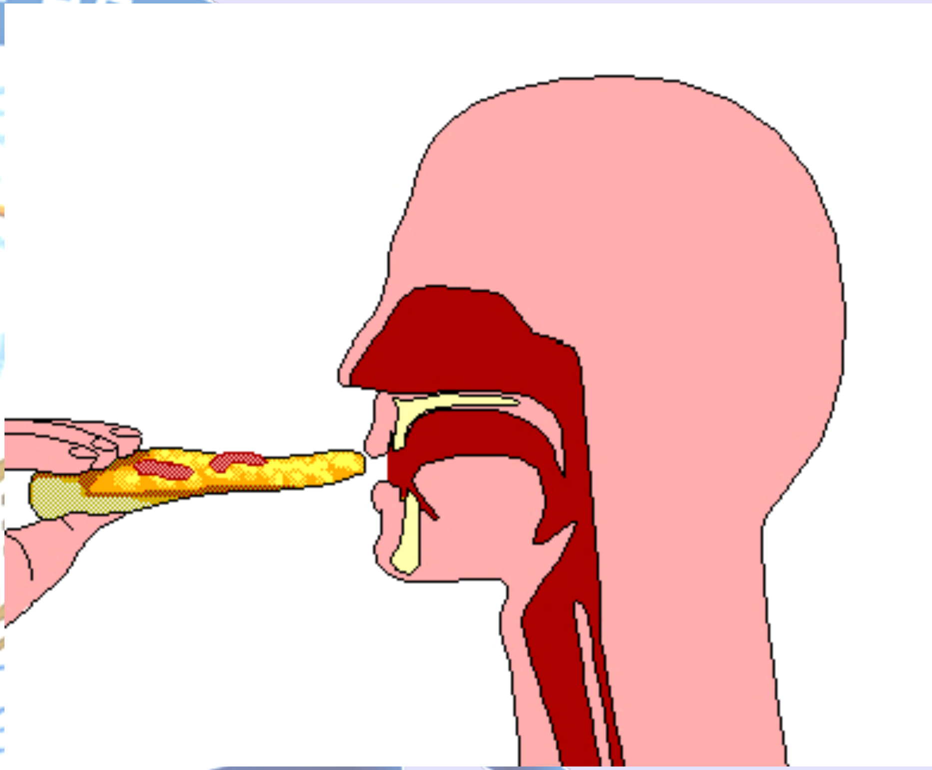
Innervation

- The tensor veli palatini is innervated by the **medial pterygoid nerve**, a branch of mandibular nerve, via otic ganglion..



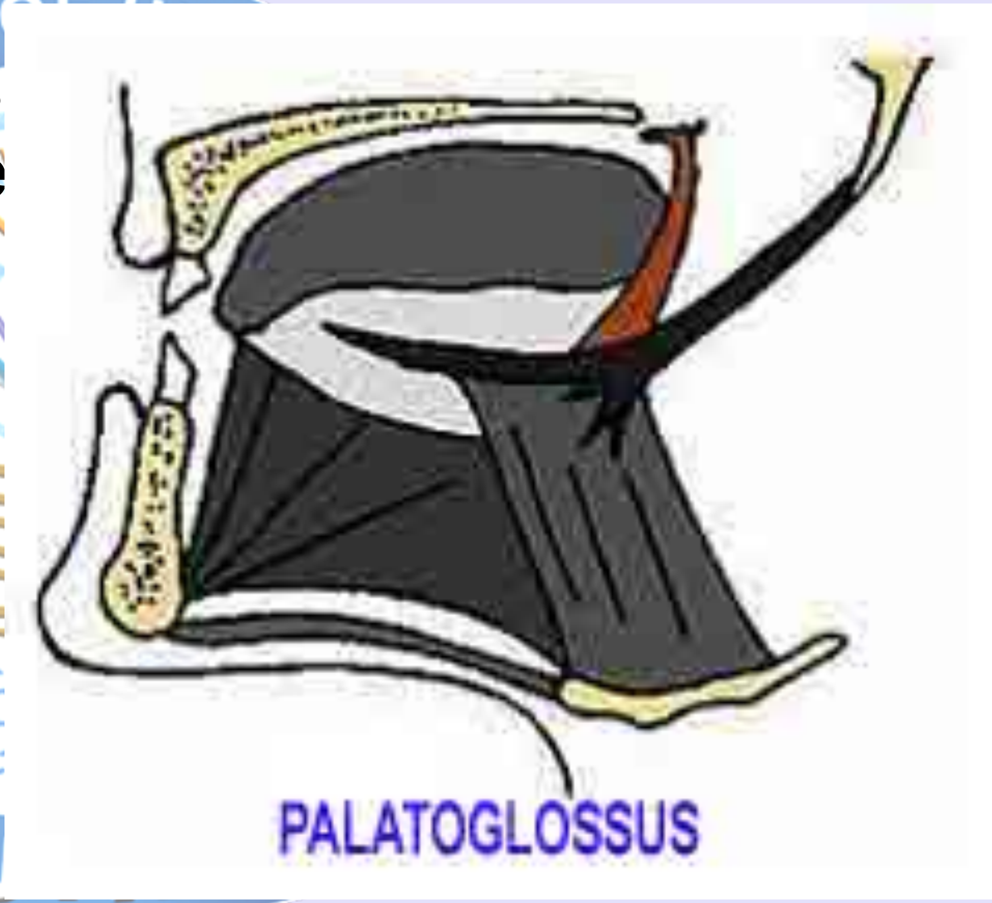
Actions

- Tenses the soft palate and by doing so, assists the levator veli palatini in elevating the palate to occlude and prevent entry of food into the nasopharynx during swallowing. **Deglutition.**



3. Palatoglossus

A small fleshy fasciculus narrower in the middle than at either end, forming, with the mucous membrane covering its surface, the glossopalatine arch.



Origin And Insertion Palatoglossus

- Arises from anterior surface of soft palate,
- Inserted into the side of the tongue, some of its fibers spreading over the dorsum, and others passing deeply into the substance of the organ to intermingle with the transversus linguae.

Palatoglossus muscle

Innervation: **Pharyngeal branch of vagus nerve via pharyngeal plexus..**

Action

- Elevates posterior tongue, closes the oropharyngeal isthmus, and aids initiation of swallowing.

Palatopharyngeus

A long, fleshy fasciculus, narrower in the middle than at either end, forming, with the mucous membrane covering its surface, the palatopharyngeal arch.

Attachment

Superior Attachment:

- Hard palate and palatine aponeurosis.

Inferior Attachment:

- Lateral wall of pharynx.

Innervation and Action

- **Innervation:**

  Pharyngeal branch of vagus nerve via pharyngeal plexus.

Action:

Tenses soft palate and pulls walls of pharynx superiorly , anteriorly, and medially during swallowing.

Musculus Uvula.

Origin and insertion

- It arises from the posterior nasal spine of the palatine bones and from the palatine aponeurosis.
- It descends to be inserted into the uvula.

Innervation and Action

Innervation:

- It is innervated by the pharyngeal branch of the vagus nerve via the pharyngeal plexus.

Action:

- functions to move and shape it.

Cleft

- In the birth defect called cleft palate, the left and right portions of hard plate are not joined, forming a gap between the mouth and nasal passage.

thanks